

BMW Group Standard

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Replacement for
GS 98003:2021-02

FMEA

Failure Modes and Effects Analysis

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Foreword

The German version is binding.

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This Group Standard has been coordinated with the responsible departments of the BMW Group. The names of the persons creating, releasing and checking this Group Standard are filed and can be retrieved in the master data of the TEREK system.

Amendments

The following amendments have been made to GS 98003:2021-02:

- section 3 updated;
- sections 3.3 and 4 added;
- sections 4.2, 4.4, 4.5, 4.6, 4.7 and 4.8 updated, are now sections 5.2, 5.4, 5.5, 5.6, 5.7 and 5.8; regards 5.6: The risk assessment is based on the action prioritization (AP) of VDA/AIAG with a shift of the boundary between low and medium risk towards low E evaluations (so far: evaluation according to B*A/B*E). The A, E evaluations according to VDA/AIAG handbook apply as well, the evaluation table for A, E is omitted.
- sections 5.2, 5.3, 5.5, 5.6 and 5.8 updated, are now sections 6.2, 6.3, 6.5, 6.6 and 6.8;
- sections 7 and 8 added;
- annex A is now annex C
- annex A „Timeline“ added
- annex B „Quality checklist“ deleted, converted into separate document (in GDMS).
- annex B „Examples for Significance“ added
- Normative references updated;
- Group Standard editorially revised.

Previous editions

2021-02

1 Scope and purpose

This Group Standard regulates the creation of design and process FMEAs within the BMW Group and its suppliers and supplements the AIAG-VDA FMEA Manual. This Group Standard is only valid in connection with the AIAG-VDA FMEA Handbook as governing basis for the application of the FMEA method.

Regulations of specialist departments and suppliers shall meet the minimum requirement of this Group Standard.

NOTE The FMEA is a qualitative, preventive method for determining and evaluating potential defects and failure states of a system, a subsystem, a part or a manufacturing process, for defining risk-based actions and documenting their effectiveness.

2 Normative references

This Group Standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

GS 91011	Special Characteristics and L-Marking; Identification, designation, and handling of Special Characteristics and L-marking
GS 98000	Statistical capability studies
AIAG-VDA FMEA-Handbook	Failure Mode and Effects Analysis; Design FMEA, Process FMEA, Supplemental FMEA for Monitoring & System Response
DID-DE-0233809 ¹⁾	FMEA Checklist Quality

1) available in Group DMS or in IPQ in the "Generic tools" menu

3 Abbreviations, terms and definitions

3.1 Abbreviations

AP	Action priority
BBG	Confirmation build phase
BM	Special characteristic
BtP	Build-to-print
D	Design
DFMEA	Design FMEA
DOC	Direct Ordering Confirmation
FMEA	Failure Mode and Effects Analysis
gAMS	Change management incl. approval
GDMS	Group Document Management System
ItO	Idea to offer
LH	Requirements specification
LSV	Service Interface Agreement
MT	Module carrier
PFMEA	Process FMEA
PQM	Product Quality Management
PROF	Production release
PZS	Production trial run
RG	Degree of maturity
RPZ	Risk priority number
SOP	Start Of Production
SW	Software
VERF	Test release
VORF	Pre-release
VS0	Pre-series 0
VS1	Pre-series 1
VS2	Pre-series 2

3.2 Terms

See AIAG-AIAG-VDA FMEA Handbook, section "Terms".

3.3 Definitions

There are no definitions in this document.

4 Designation

There is no designation in this document.

5 Design FMEA step-by-step approach

5.1 General information

The DFMEA is performed in seven steps.

NOTE Due to the interactions between the steps, it may be necessary to implement changes in steps that have already been carried out and thus to obtain analysis results in part iteratively.

5.2 Planning and preparation

Supplement to the AIAG-VDA FMEA Handbook, section "Design FMEA Step 1: Planning and preparation"

A boundary diagram (see AIAG-VDA FMEA Handbook, "Block/boundary diagram") of the analysis object(s) shall be created to define the scope of consideration of a design FMEA (DFMEA).

Customer operating states and system states of the scope of consideration shall be included.

NOTE 1 The responsibilities for interfaces to corresponding systems shall be clarified; proper coordination is a mandatory constituent of planning.

The DFMEA shall exist in German or English. It shall be ensured that all affected persons in all relevant business units understand the FMEA logic and are able to process it.

The analysis object shall be processed using a software tool. The software tool shall exhibit

- at least three structural levels,
- function networks, and
- failure networks.

The software tool with which the FMEA is carried out shall be coordinated with and be confirmed by BMW.

To ensure the consistency of the method (and the transparency of the results), the

- scope of consideration,
- interfaces,
- the organization,
- the implementation, and
- the milestones incl. review deadlines

shall be coordinated, according to AIAG-VDA FMEA manual, section "Project planning" 5Z. In particular, the handling of directed parts, if they exist, shall be defined. Milestones shall be defined for the DFMEA reviews. The DFMEA reviews shall be implemented and documented using the FMEA quality checklist (DID-DE-0233809).

The milestones to be achieved, and the required content are described in Annex A.

Time-critical and content-related high risks shall be prioritized. The order of the design FMEA processing shall be based on a risk-oriented prioritization (e.g. known?, proven?, comparable?).

The planning and preparation results shall be documented in a profile and retained together with the DFMEA for the required legal retention periods.

The supplier shall present its procedure for creating the DFMEA based on an example and coordinate the procedure with BMW; documentation is via the FMEA checklist. The coordination shall take place in a joint design FMEA kickoff workshop between the supplier and BMW. All design FMEA moderators, project management, and technical experts shall participate. Within the scope of the workshop the profile shall be agreed and approved in a binding manner and the DFMEA be nominated.

BMW shall be granted access to the status of the DFMEA and the accompanying documents at any time. The supplier shall demonstrate how the quality of the design FMEA of his sub-suppliers and conformity to this Group Standard is ensured.

The completeness and consistency of the function and failure networks and the associated severity (S-evaluation) shall be ensured across all system levels.

NOTE 2 Completeness and consistency usually comprise several FMEAs.

5.3 Structure analysis

Supplement to the AIAG-VDA FMEA Handbook, section "Design FMEA Step 2: Structure analysis"

As part of the DFMEA the party responsible for the design (e.g. supplier) shall analyze all subsystems, components and design features.

If the scope of consideration is adjusted due to new findings from the structure analysis, the profile and the boundary diagram from the analysis and planning phase shall be updated and reconciled with the higher-level system. This applies to all steps of the FMEA methodology.

A structure tree shall be prepared for every design FMEA. This structure tree shall be function-oriented. Every structure element shall be clearly identified. The top level in the structure tree shall correspond to the designation in the design FMEA of the higher-level system. If available, part numbers and their designation shall be assigned to the structure element as an attribute.

The characteristics level shall contain geometric characteristics, characteristic material values, and design-bound manufacturing processes, and shall be represented as a separate structure level.

The completeness of the scope of consideration determined in the planning and preparation step shall be proven by means of a clear reference to documents such as parts lists, wiring diagrams, software function blocks or models.

For mechatronic systems, SW functionalities shall be analyzed in corresponding system elements.

NOTE The SW functionality may be implemented above the ECU level (e.g. as structure element), the structure element ECU then describes only the hardware function. Faults resulting from programming or software implementation are generally not considered here.

5.4 Function analysis

Supplement to the AIAG-VDA FMEA Handbook, section "Design FMEA Step 3: Function analysis"

For each structure element, all functions shall be determined and linked in function networks. All functions within the scope of responsibility of the designing party (e.g. supplier) shall be analyzed in their system states.

The findings from the "block / boundary diagram" shall be considered to analyze the complete function of interfaces.

The functions of the top level of the structure tree shall be coordinated with the design FMEA of the higher-level system.

Each function shall be described individually to be able to clearly define the function networking. The corresponding requirement parameters shall be named or referenced (e.g. requirements specification) for each function/feature.

An "automatic" linking of functions using FMEA software is not permitted.

The function description shall be clear, comprehensible, verifiable, and capable of validation.

5.5 Failure analysis

Supplement to the AIAG-VDA FMEA Handbook, section "Design FMEA Step 4: Failure analysis"

Within the scope of the failure analysis, all structure elements with all functions shall be investigated. All potential failures (failure type, failure cause, and failure effect) shall be described clearly and comprehensibly as a loss or restriction of a function or its parameters. Concept-related failures shall be investigated as well. If possible, a reference to the formulation of the function shall be established during the formulation.

A comparison with field experiences from previous projects shall be carried out and documents.

NOTE If necessary, external specialists shall be involved in the analysis of failures and actions.

For each failure mode of a structure element, the failure chain with failure causes, failure effects, and failure mode shall be created and logically linked in hierarchical failure networks.

The failures of the top level of the scope of consideration shall be compared with the higher-level design FMEA level.

5.6 Risk analysis

5.6.1 General information

Supplement to the AIAG-VDA FMEA Handbook, section "Design FMEA Step 5: Risk analysis"

The evaluation is based on the predicted series status and the current action status.

actions with planning status but not yet implemented shall have responsible persons and deadlines assigned to them.

NOTE For completed actions of the initial status no responsibilities and deadlines shall be defined. If it is ensured that feedback of all standard actions, e.g. endurance testing, into the FMEA is guaranteed, this action may be defined as the "initial action status" without deadlines and without persons responsible.

5.6.2 Severity S

Supplement to the AIAG-VDA FMEA Handbook, section "Severity (S)" in section "Design FMEA Step 5: Risk analysis"

The evaluation of the severity of a failure is generically based on the most critical failure consequence with respect to the vehicle. The severity scale according to the AIAG-VDA FMEA Handbook, table "DFMEA severity (S)" shall be applied and represents a minimum standard. Refer also the translation table "BMW BI to S in Annex B.

Higher classifications may be applied if this is necessary from the customer or company perspective (e.g., visual/tactile requirements in the interior).

The severity evaluation of the failure effect in the top element of the structure tree shall be copied from the superordinate system level or the higher-level DFMEA.

The severity of the failure effect shall be agreed between BMW and the supplier, whereby the initial severity evaluation is specified by BMW. This also applies to sub-suppliers of all TIER levels.

In the case of safety-critical failure effects, the corresponding specialist departments shall be involved in the evaluation. These specialist departments ensure that the controllability of the product in the event of a failure and the relevance of the operating states are considered in the evaluation. The decisions shall be documented.

5.6.3 Risk assessment: Action prioritization

The prioritization shall be in line with the following logic, with a more stringent prioritization compared to the AIAG/VDA Handbook, see table Risk Prioritization Logic.

The occurrence and detection shall be evaluated in accordance with AIAG/VDA, taking into account findings from the field.

The risk assessment thus differs fundamentally from the previous version of this Group Standard (GS 98003:2021-02). The assessment logic on which the FMEA is based (O, D, risk matrix) shall be documented. Assessment O, D, and the risk matrix in a FMEA shall be based on a logic. If that is not possible, the subdomains of a FMEA shall follow a consistent logic and be assessed separately.

The use of the standard matrices S*O/S*D or AP from the AIAG-VDA FMEA manual or prioritization according to RPN are not permitted.

Table Action Prioritization (AP) logic BMW Design

Figure 1 shows the action priorities high/medium/low in the colors red/yellow/green in the matrices O*D in ascending order of severity from left to right.

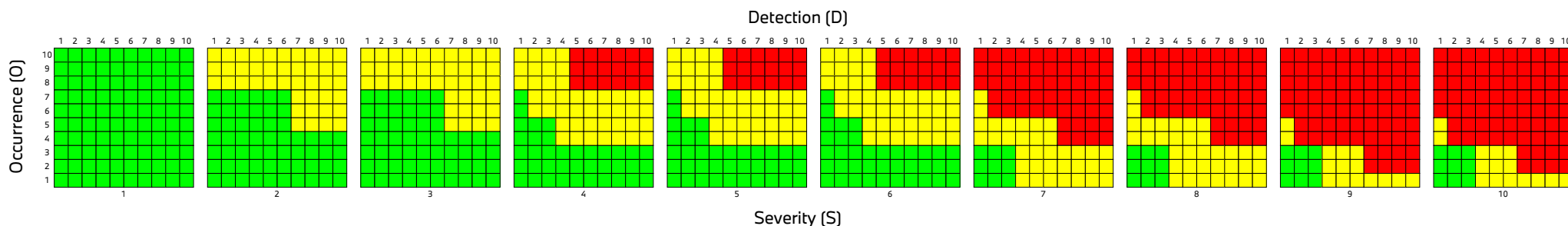


Figure 1: Matrices O*D with severity in ascending order

NOTE The transformed logic in Figure 2 is the equivalent representation as used in some programs.

		Detection				
Severity	Occurrence	1	2 to 3	4	5 to 6	7 to 10
1	1 to 10					
2 to 3	1 to 4					
	5 to 7					
	8 to 10					
4 to 6	1 to 3					
	4 to 5					
	6 to 7					
	8 to 10					
7 to 8	1 to 3					
	4 to 5					
	6 to 7					
	8 to 10					
9 to 10	1					
	2 to 3					
	4 to 5					
	6 to 10					

Figure 2: Transformed view

5.7 Optimization

5.7.1 General information

Supplement to the AIAG-VDA FMEA Handbook, section "Design FMEA Step 6: Optimization"

Based on the BMW action prioritization actions shall be defined that transfer the action priority to the low level.

NOTE For this purpose, it may be necessary to define the actions together with the superordinate decision-making level in the project.

After completing the action, this forecast shall be reassessed in conjunction with all other actions and documented with reference to appropriate evidence (e.g. gAMS).

Optimization is complete when all risks with low action priority are assessed and completed or, for all medium risks with safety or legal implications and all high risks a documented release with argumentation has been submitted to the next higher management level.

Actions may be assigned to low evaluations in order to improve quality.

5.7.2 Occurrence O:

Supplement to the AIAG-VDA FMEA Handbook, section "Occurrence (O)" in section "Design FMEA Step 5: Risk analysis"

In DFMEA Step 6 "Optimization", the potential for the occurrence of the failure cause in customer operation shall be evaluated taking into account all available findings and avoidance actions. New findings shall be considered in a new evaluation.

5.7.3 Detection D:

Deviation from the AIAG-VDA FMEA Handbook, section "Detection (D)" in section "Design FMEA Step 5: Risk analysis"

After the implementation of the detection action, the effectiveness of the detection action shall be checked and newly evaluated. If the results of the detection action do not match the prognosis of occurrence, the occurrence shall be newly evaluated.

5.8 Results documentation

Supplement to the AIAG-VDA FMEA Handbook, section "Design FMEA Step 7: Results documentation"

The participants and the partial result achieved shall be documented for each DFMEA meeting.

The DFMEA report for the results documentation according to AIAG-VDA FMEA Handbook "FMEA documentation" includes at least:

- the target/actual comparison with the original profile;
- the block/boundary diagram;
- the risk matrix BMW AP;
- the procedure for resolving open actions and high risks;
- the description of the timing sequence for continuous improvement of the DFMEA (as described in the AIAG-VDA FMEA Handbook);
- the consistency with primary and secondary DFMEA, e.g. S-evaluations, functions, special characteristics.

Documentation shall be created for the requirements specification released for development, the production tool release, and the production release; this applies internally at BMW and for suppliers, if affected.

The documentation and the FMEA shall be retained in accordance with the specified legal retention periods. If the DFMEA is relevant for a release, classification shall follow BMW Group instructions „Management and retention of documents" Annex „Classification system for documents, adapted for BMW AG Deutschland", class number 5.3.

6 Process FMEA step-by-step approach

6.1 General information

The regulations for the PFMEA described in this document are binding for all PFMEA used for process release.

6.2 Planning and preparation

Supplement to the AIAG-VDA FMEA Handbook, section "Process FMEA Step 1: Planning and preparation"

All processes that influence product quality shall be analyzed to determine risk-based economic validation actions.

To determine the scope of consideration of a PFMEA, the full value creation between goods receipt and goods issue shall be considered in accordance with the responsibility model. In doing so, it shall be determined which scopes are not to be considered in the PFMEA and how the interfaces between the PFMEA are coordinated.

This definition of the scope of consideration may include not only the manufacturing processes but also relevant sections of the logistics processes or interfaces to these. It shall be decided which rework, repair, test, bypass, and backup processes shall be included as a mandatory constituent of the scope of consideration.

The pertinent processes of all suppliers (including system/machinery suppliers and sub-suppliers) shall also be considered here.

Responsibilities shall be clarified for interfaces to corresponding systems. Appropriate coordination is a mandatory constituent of the planning.

The PFMEA shall be prepared in German or English. It shall be ensured that all affected persons in all relevant business units understand the FMEA logic and are able to process it.

The analysis object shall be processed using a software tool. The software tool shall exhibit

- at least three structural levels,
- function networks, and
- failure networks.

The software tool with which the FMEA is carried out shall be coordinated with and confirmed by the respective BMW quality departments.

The planning and preparation results shall be documented in a profile and retained together with the PFMEA for the required legal retention periods.

To ensure the consistency of the method at the (system) supplier (and the transparency of the results),

- the scope of consideration,
- the interfaces,
- the organization,
- the implementation, and
- the milestones incl. review deadlines

shall be defined along with the BMW contact persons.

The profile is used as communication and documentation aid. The PFMEA reviews shall be implemented and documented using the FMEA quality checklist (DID-DE-0233809).

The milestones to be achieved, and the content required for this, are described in Annex A.

Coordination between the supplier and BMW shall take place at a joint PFMEA kickoff workshop. All PFMEA moderators, project management, and technical experts shall participate. Within the scope of the workshop, the profile shall be agreed and approved in a binding manner, and the PFMEA be nominated.

BMW may be granted access to the status of the supplier's and the sub-supplier's PFMEA at any time. The supplier shall demonstrate how the quality of the PFMEA of his sub-suppliers and conformity to this Group Standard is ensured.

6.3 Structure analysis

Supplement to the AIAG-VDA FMEA Handbook, section "Process FMEA Step 2: Structure analysis"

A structure tree shall be prepared for every PFMEA. This tree is based on the process flow diagram and the material flow diagram.

The top level of the structure tree is the overall process.

An intermediate level for the process steps is necessary. For these process steps, the work content shall be appropriately detailed on the basis of the work plan.

NOTE Similar work contents may be combined in the same process step (e.g. bolting).

The relevant cause elements (e.g. measurement, human, machine, material, environment, method) shall be named at the lowest level of the structure tree.

6.4 Function analysis

Supplement to the AIAG-VDA FMEA Handbook, section "Process FMEA Step 3: Function analysis"

The production functions of the top process element of the structure tree shall be coordinated between the supplier and the BMW contact person.

Product characteristics and process characteristics shall be assigned to structure elements. It is not necessary to describe tolerances in the PFMEA if these become obvious from referenced documents (e.g., work instructions, drawing).

6.5 Failure analysis

Supplement to the AIAG-VDA FMEA Handbook, section "Process FMEA Step 4: Failure analysis"

During the failure analysis, the focus element is either the process step or an individual work content of the process step.

A failure chain with the failure cause, failure effect, and failure mode shall be created.

The failure causes shall describe the root cause of the failure mode. Next, it shall be checked, whether the failure causes may be combined in a reasonable way.

The following basic assumptions are permissible during the determination of potential failure causes:

- Incoming parts and materials meet the relevant specifications.

Failure effects include the effects from the viewpoint of:

- the final customer;
- the legal/regulatory requirements;
- the plant supplied;
- one's own plant.

The critical failures from the viewpoint of the customer of the DFME (system/subsystem or component) shall be reconciled with the failure effects of the process FMEA.

6.6 Risk analysis

Supplement to the AIAG-VDA FMEA Handbook, section "Process FMEA Step 5: Risk analysis"

The initial status of the actions shall include all physically available detection and avoidance actions based on evidence of suitability. Responsibilities and deadlines shall not be defined for completed actions of the initial status.

Planned actions shall be recorded as separate action status.

NOTE Higher classifications may be used for the S-, O-, and D-evaluations if this is necessary from the customer or company perspective (e.g., visual/tactile requirements in the interior).

Supplement to the AIAG-VDA FMEA Handbook, table "PFMEA Occurrence (O)"

Table 1 applies to the risk classification.

EXAMPLE "Severity 9 to 10" + "Occurrence 4 to 5" + "Detection 7 to 10" $\triangleq$ high priority of the action for risk reduction

Table 1: Risk classification

Severity	Occurrence	Detection			
		1	2 to 4	5 to 6	7 to 10
1	1 to 10	L	L	L	L
2 to 10	1	L	L	L	L
2 to 3	2 to 7	L	L	L	L
	8 to 10	L	L	M	M
4 to 6	2 to 3	L	L	L	L
	4 to 5	L	L	L	M
	6 to 7	L	M	M	M
	8 to 10	M	M	H	H
7 to 8	2 to 3	L	L	M	M
	4 to 5	M	M	M	H
	6 to 7	M	H	H	H
	8 to 10	H	H	H	H
9 to 10	2 to 3	L	L	M	H
	4 to 5	M	H	H	H
	6 to 10	H	H	H	H
H high prioritization of the risk reduction actions M medium prioritization of the risk reduction actions L low prioritization of the risk reduction actions					

6.7 Optimization

No supplement to the AIAG-VDA FMEA Handbook, section "FMEA Step 6: Optimization"

Optimization is complete when all risks with low action priority are assessed and completed or, for all medium risks with safety or legal implications (exception E = 1) and all high risks a documented release with argumentation has been submitted to the next higher management level.

Actions may be assigned to low evaluations to improve quality.

6.8 Documentation of FMEA results

Supplement to the AIAG-VDA FMEA Handbook, section "FMEA Step 7: Documentation of FMEA results"

The participants and the partial result achieved shall be documented for each PFMEA meeting.

The PFMEA report for the results documentation according to AIAG-VDA FMEA Handbook "FMEA documentation" includes at least:

- the target/actual comparison with the original profile;
- the process flow sequence (priority graph);
- the risk matrices AP;
- the procedure for completing open actions;
- a description of the chronology for continuous improvement of the PFMEA (as described in the AIAG-VDA FMEA Handbook);
- consistency with higher-level DFMEA:
 - S-assessments and
 - functions at least for special characteristics.

The documentation shall be created for the release of the production tool and for the production release.

The documentation and the FMEA shall be retained in accordance with the specified legal retention periods. Classification of PFMEA according to BMW Group instructions „Management and retention of documents“ Annex „Classification system for documents, adapted for BMW AG Deutschland“ with class number 7.2.

7 Living FMEA

Supplement to the AIAG-VDA FMEA Handbook

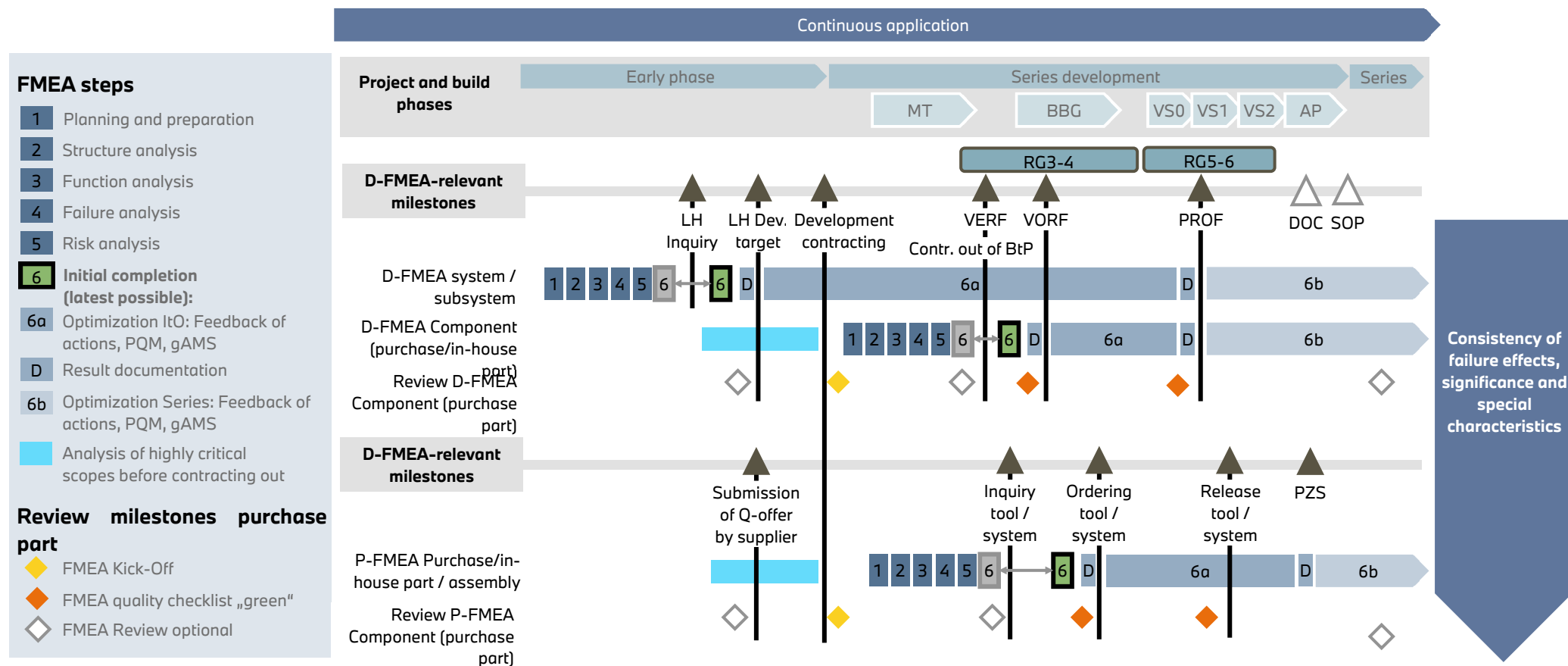
Each FMEA shall be updated during the entire product service life with new relevant findings from a system, design, and process perspective (including change management and problem management). Attention shall be paid to the consistency of the DFMEA at all system levels and areas of responsibility. Adaptations of the FMEA due to product/process problems shall be documented in the 8D report or equivalent report/tool).

8 Special characteristics

Deviation from the AIAG-VDA FMEA Handbook, annex "Special characteristics"

GS 91011 shall be applied.

A (Annex normative) schedule



Additional notes:

- The generic scheme of the relation between the FMEA types during the development phase may deviate from the individual boundary conditions of the project.
- At the time of release of the requirements specification for the development target (E-Ziel LH) the first optimization of the D-FMEA / subsystem has been completed, the initial status of the special characteristics and failure effects been harmonized and the quality checklist is "green".
- Upon pre-release actions shall be assigned to all high risks of all types of D-FMEA.
- Upon series tool release actions shall be assigned to all high risks of the P-FMEA.

Figure A.1: Timeline

B (Annex informative) Severity

Figure B. 1 shows a translation of the BMW internal BI-rating to the FMEA S rating including examples with partly different specificities. The assignment BI to S may vary for comfort functions ($S \leq 6$; $BI \geq 4$) by ± 1 .

Coarse cluster	Evaluation of the possible failure effects according to AIAG			Assignment to BMW BI rating			Ramp-up relevance	Examples	Examples with variable characteristics	
	S	Effect	Criteria of Severity (S)	BI	BI rating concept	BI rating failure		Examples	Oil loss	Acoustics
Legal and safety	10	very high	Impact on the safe operation of the vehicle and/or other vehicles, the health of the driver or the front passenger, or other road users.	1	Veh. / tech. solution has safety defects	Product safety I, failure of safety-relevant systems,	preventing	Brake system failure; undesirable acceleration/braking $\leq 1,5 \text{ m/s}^2$; sudden loss of propulsion force; extended blocking of a wheel ($v = 0$) of an axle at high speed; Convertible top opens unrequested or due to foreseeable faulty operation at high speeds	Risk of fire due to drips	
	9		Noncompliance with legal requirements and requirements from authorities		Veh. / tech. solution does not meet; conformity-relevant requirements	Product conformity affected, approval-relevant component affected	preventing	Legal requirement for homologation not met (e.g. type with compulsory L-marking according to GS 91011 not available, continuous output (ECE R85) not available, ETK part number (Emission Related Part / Component US); General legal requirements (e.g. OBD, EMV emissions limits, material requirements (GADL, IMDS), Technical Vehicle Regulations (FATEV), normative requirements)	Environmental pollution on a larger scale (e.g. breakage of transmission, no drips)	
loss / impairment of vehicle operation	8	high	Loss of a major function required for normal vehicle operation over the intended service life	2	Veh. / tech. solution causes extreme annoyance	failure / breakdown to be expected, threat of legal action / publication	preventing	Red warning lamp, acceleration too slow, preventing appropriate driving for customer, vehicle fails to unlock		Noises that usually lead to elimination of the damage
	7		Restriction of a major function required for normal vehicle operation over the intended service life	3	Veh. / tech. solution causes annoyance, leads to a negative purchase decision and negative media feedback	Immediate workshop visit	preventing	Driving function restricted, yellow warning lamp, convertible top fails to close		
Warranty cases	6	medium	Loss of a comfort function	4	Veh. / tech. solution faulty, leads to disappointment, customer rejection or negative media feedback	Defect leads to an unscheduled workshop visit	preventing	Convertible top fails to open, Loss of entertainment function		Noises that indicate damage
	5		Restriction of a comfort function				preventing	Restricted radio reception		
	4		Clearly perceptible quality impairment of look, sound, vibrations, roughness or tactile experience	5	Veh. / tech. solution leads to dissatisfaction, difficulty in making purchase decision or critical media feedback	Defect is complained of during the next scheduled workshop visit	preventing	Visible surface corrosion without functional effect	Visual impairment (oil leak)	Annoying noises (slight rattle, creaking, squeaking)
Not premium	3	low	Moderately perceptible quality impairment of look, sound, vibrations, roughness or tactile experience	6	Veh. / tech. solution does not fully meet expectations, is only mediocre	Error is noticed by the customer and causes displeasure	hazardous			Borderline NVH that does (normally) not cause warranty cases / costs
	2		Marginally perceptible quality impairment of look, sound, vibrations, roughness or tactile experience	7	Veh. / tech. solution good, irritating in individual / rare cases / situations / behaviors	Minor deviations from desired condition are identifiable	%			
	1	very low	No perceptible effect	8	Veh. / tech. solution is very good and causes satisfaction, expectations are fully met	100 % free of defects				
	1	very low	No perceptible effect	9	Veh. / tech. solution is excellent, the technical solutions are convincing					
	1	very low	No perceptible effect	10	Veh. / tech. solution is excellent, the technical solutions are impressive					

Figure B. 1: Translation of the internal BMW BI-rating to the FMEA S rating

C (Annex normative) Summary Table C.1 PFMEA

Table C.1: Summary table

		AIAG-VDA FMEA Handbook Table "PFMEA Occurrence (O)"	AIAG-VDA FMEA Handbook Table "PFMEA Occurrence (O)"	VDA FMEA Handbook Table "Alternative table PFMEA Occurrence (O) with incidents per 1 000 cases/vehicle"	VDA FMEA Handbook Table "Alternative table PFMEA Occurrence (O) with frequency forecast of the failure cause"
Occurrence		Type of prevention	Avoidance actions	Incidents	Frequency prognosis of the failure cause
10	extremely high	none	No avoidance actions	> 100 000 ppm	Always
9	very high	Behavior	Avoidance actions have little effect in preventing the failure cause	> 50 000 ppm	Almost always
8				> 20 000 ppm	more than 1 x per shift
7	high	Behavior or technically	Avoidance actions have medium effect in preventing the failure cause	> 10 000 ppm	more than 1 x per day
6				> 2 000 ppm	more than 1x per week
5	medium		Avoidance actions are effective in the prevention of the failure cause	> 500 ppm	more than 1 x per month
4				> 100 ppm	more than 1 x per year
3	low	Proven procedures: Behavior or technically	Avoidance actions are highly effective in the prevention of the failure cause	> 10 ppm	1x per week
2	very low			> 1 ppm	less than 1 x per year
1		technical	Poka-Yoke action with proven effectiveness. Production of defective products ruled out due to design (component geometry) or process (appliance or tool design). Effective proofing against inappropriate use has been demonstrated on the product. Type of failure avoided by design or process or detection methods (provably) always detect the type of failure or cause of failure.	The failure is eliminated by the avoidance action.	never